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A new state-of-health estimation method for lithium-ion batteries through the intrinsic relationship between ohmic internal resistance and capacity

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ABSTRACT

For secure and reliable operation of lithium-ion batteries in electric vehicles, diagnosis of the battery degradation is essential. This can be achieved by monitoring the increase of the internal resistance of the battery cells over the whole lifetime of the battery. In this paper, a method to estimate state of health (SoH) is presented through the established linear relationship between ohmic internal resistance and capacity fade. Firstly, the Thevenin model and the recursive least squares (RLS) algorithm are applied to simulate battery dynamic characteristics and identify model parameters, respectively. Secondly, based on the established linear relationship between ohmic internal resistance and capacity fade, both ohmic internal resistances at the start and the end of the battery's lifetime are estimated by only two random discharge cycles at different aging stages. Finally, an online SoH estimator is formulated and applied to estimate the SoH of a battery's remaining cycles. In addition, a series of experiments were carried out based on dynamic loading to verify the proposed method. The SoH estimates indicate that the evaluated maximum SoH errors are within ± 4%. The proposed SoH estimation method is consistent with the measurement data of the battery and shows good results with very low computational effort.

1. Introduction

To address the challenges associated with increasing world energy consumption and climate change, electric vehicles (EVs) can play a key role. The development of suitable power batteries in EVs is essential, and focuses on the lithium-ion battery due to its excellent power characteristics [1]. Being the main power source of EVs, it is important to manage the battery to extend its lifespan, improve its reliability, and lower its cost. In order to guarantee the realization of these functions, a battery management system (BMS) is necessary [2]. As the main part of the energy system, the BMS plays a crucial role for states estimation, battery monitoring, cell balancing, thermal management, and others [1,3,4]. One of the core functions of state estimation is real-time SoH estimation [3]. SoH estimation is very important for guaranteeing the system's performance and reliable operation, and it significantly affects the overall vehicle performance and life cycle [5].

Conventionally, SoH is a parameter, which indirectly represents the age condition of the battery cell and the remaining lifespan [6–8]. It mainly reflects in two aspects: capacity fade and power fade [6,9,10]. Hence, the SoH can be estimated using the capacity or the impedance of the battery.

A wide variety of SoH estimation methods [9,11] have previously been summarized, each one has its own advantages and disadvantages. The SoH estimations mainly include three categories [8]: (1) direct measurement method; (2) model-based method; (3) data-driven method.

(1) *Direct measurement method:* From the perspective of capacity, the present maximum available capacity can be obtained using the open-circuit voltage (OCV) method and the Coulomb-counting method. From the perspective of impedance, the impedance of the battery can also be measured with specialized equipment [12,13]. The OCV method can determine the battery capacity using the relationship between OCV and state of charge (SoC). And the variation of OCV model is small over the battery lifetime. Thus, the accuracy of the method for aged batteries is almost as high as those new battery [1]. However, the method requires a long rest-time to measure OCV accurately until the battery reaches a steady state [14]. The Coulomb-counting method can determine battery capacity by integrating the battery discharge current, but this is time consuming [8] and an accumulated error and sensor noise represent significant problems. Electrochemical impedance spectroscopy

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(EIS) can determine the impedance using impedance spectroscopy analysis [9]. Resistance test equipment and the Joule effect can also determine the internal resistance [9]. In summary, a direct measurement method usually has less computational complexity and is possible to implement in a BMS. The drawbacks are that it has a low accuracy, and usually requires special equipment and is not suited for situ estimation [1,9].

(2) *Model-based method:* The main idea of the model-based SoH estimation is to connect the measured battery signals (voltage, current, and temperature) with the battery SoH employing a battery model [1]. Adaptive models are developed to describe the dynamic characteristics of the lithium-ion battery, including electrical and electrochemical models. Adaptive techniques known from control theory are employed with these adaptive models. For instance, in Ref. [6], the SoH and SoC are estimated by the Extended Kalman filter (EKF), but the SoH is updated offline and the time scale of the SoH is based on model accuracy. In Ref. [15], the least squares technique is used to estimate SoH. However, the present methodology should be validated at different C-rates and temperatures for a more realistic performance. In Ref. [16], the particle filter (PF) is employed to estimate both the capacity and SoC, but the accuracy is limited because of the simplified battery model. These methods are online and closed-loop [8], and their performance has been validated with different experiments. However, the accuracy and robustness of the model decides the accuracy of the method.

(3) *Data-driven method:* The battery is considered a "black box" to estimate battery SoH without battery model in these methods. For instance, in Refs. [17,18], artificial neural network (ANN) is applied to estimate SoH. The support vector machine (SVM) is used for SoH estimation in Refs. [19,20]. Besides, DVA (Differential Voltage Analysis) [21] and ICA (Incremental Capacity Analysis) [2] are also developed to estimate SoH. The framework of such methods consists of off-line training and online SoH estimation [18]. In this process, the feature extraction of battery degradation is also necessary. What's more, a precise battery model is not necessary for these data-driven methods, but these methods require a large database for different working conditions during the lifetime. And the DVA and ICA method require a constant and low current to discharge the battery in order to acquire the accurate DV curves or IC curves. In addition, these methods often require high computing power and they may not be suitable for BMS, which only contains some microcontrollers or low-cost systems [22]. Furthermore, the battery parameters may change with the aging process in practical applications.

The values of battery capacity usually are nonlinear, time varying, and uncertain. Thus, it is still difficult to estimate the capacity of a battery precisely. Lithium-ion battery degradation occurs mainly because the active substances, lithium ions and electrolyte, solidify gradually in the SEI. As a result, they cannot continue to participate in the battery electrode reaction. While the concentration of lithium ions is closely linked to the capacity and the conductive properties of the battery, a loss of lithium ions will lead to the decline in capacity and the rise in impedance [23]. In addition, stressing the lithium-ion battery by overcharging, exposure to extreme temperatures, or aging changes also the carbon-based anodes irreversibly. The aging effect of the anodes may be mostly attributed to changes at the SEI [9]. This electrochemical mechanism increases the impedance and reduces the capacity at the same time, and they follow approximately the same trend together [10].

As previously shown, SoH estimation should be made by both: capacity and resistance estimation. However, the resistance is relatively simple and fast to determine. The current algorithms to identify the resistance are precise enough when they use the equivalent circuit model. And Haifeng et al. [24] proposed a new definition of SoH based on ohmic internal resistance. Hence, it is a good choice to estimate SoH via ohmic internal resistance.

In order to estimate the SoH based on the intrinsic relationship between ohmic internal resistance and capacity [25], the ohmic internal resistance of both a new battery (R_{new}) and end-of-life battery (R_{eol}) are important [1,24]. However, due to the irreversible aging process, it is difficult to acquire R_{new} when the battery has been used for some time. Also R_{eol} cannot be measured easily either until the battery is at the end of lifetime.

To solve this problem, a new method is proposed to estimate SoH by using the correlation between ohmic internal resistance and capacity. Thevenin model is suitable for modeling the relaxation effect and the dynamic behavior of lithium-ion batteries. A recursive least squares algorithm with a forgetting factor is used to realize accurate identification of the model parameters [1,6,9]. Based on the correlation between the ohmic internal resistance and capacity, a linking equation to calculate the R_{new} and R_{eol} can be built. Finally, the values for R_{new} and R_{eol} can be estimated via two discharge tests, and the SoH estimation method is developed. Different from other SoH estimation methods based on internal resistance, the linear relationship between the battery impedance increase and the battery capacity loss is built and applied through only two online resistance tests and off-line capacity tests. This method does not require complex matrix operations, special battery tests, and too much previous knowledge of the operation performance of the battery. In addition, a series of battery aging experiments at different temperatures are carried out to demonstrate the validity and precision of the proposed online estimation method.

The remainder of this paper is organized as follows: Section 2 reports our experimental settings and aging experiments of batteries. Section 3 develops the framework of our estimation method, and Section 4 elaborates and discuss the results of the propose method. Section 5 concludes the paper.

2. Experimental

The proposed method is designed for online SoH estimation. Therefore, recorded data of the dynamic load of electric vehicles are required through the whole lifecycle.

2.1. Test bench

The battery test bench is shown in Fig. 1. It consists of an ITECH electronic load IT8511+, an ITECH IT6523D as a charger, a set of batteries (which may reside either inside or outside an environmental chamber), a suite of sensors (voltage, current, and temperature), some custom switching circuitry, a data acquisition system (DAQ) and a computer for control and analysis. In addition, a battery monitoring software PV8500 is used to control the IT8511+ load.

The IT8511+ can discharge a battery/battery pack according to the program with a maximum discharge current of 30 A and the maximum

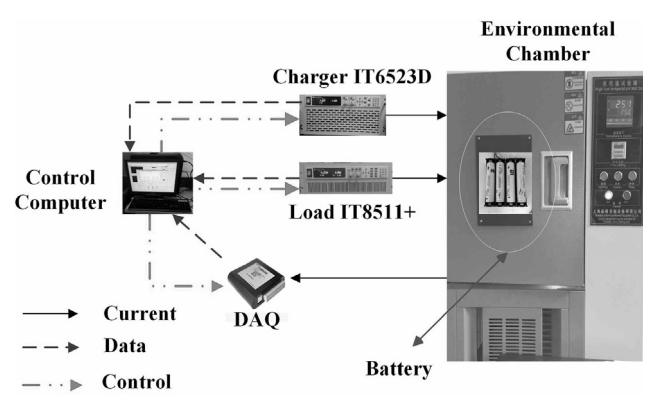

Fig. 1. The architecture of battery test bench.

Table 1
Main specifications of the LiNMC cell.

Nominal capacity (Ah)	Nominal voltage (V)	Charge cut-off voltage (V)	Discharge cut-off voltage (V)
2.60	3.70	4.20	2.75

voltage of 120 V. The error of the current measurements of IT8511+ is less than 0.1% of the measured value +0.1% I (FS – current of full scale). The error voltage measurements of IT8511+ are less than 0.1% of the measured value +0.1% U (FS – voltage of full scale). The unit of current measurement accuracy is ampere (A) and the unit of voltage measurement accuracy is volt (V). The batteries' data, including terminal voltage, load current, and discharge capacity (Ah), temperature, and other signals, can be collected and recorded during charging and discharging. The data acquisition frequency is 1 Hz, and these recorded data (terminal voltage, load current, and discharged capacity) are transmitted to the control computer via a USB cable. The battery used for this test is a lithium-nickel-manganese-cobalt (LiNMC) ICR18650-26F from Samsung with a graphite anode. Its main specifications are listed in Table 1. And the two batteries are numbered 30 and 17 (sample No. 30 and sample No. 17), respectively.

2.2. Aging experiment design

The test procedure of sample No. 30 is shown in Fig. 2. It is designed to investigate the battery's characteristics, the ohmic internal resistance, and the capacity during the whole aging process based on the specification of the battery. The experimental temperature is 35 °C.

The detailed procedure comprises following steps:

Standard charge: Charging the battery initially with constant current at 0.5 C (1.3 A) to 4.20 V and then with constant voltage at 4.20 V till charge current declines to 0.01 C (0.026 A), the charge stages are shown as Fig. 3, then the battery is kept at rest for 1 h.

Capacity test: The capacity C_{cur} is the discharge capacity of the battery, which is measured by IT8511+ with discharge current of 0.1 C (0.26 A) with 2.75 V cut-off within 1 h after the standard charge. The initial capacity is defined as C^1_{cur}, the ith capacity is C^i_{cur}. And the ith decrease of the capacity (C^i_{dec}) is defined as follows:

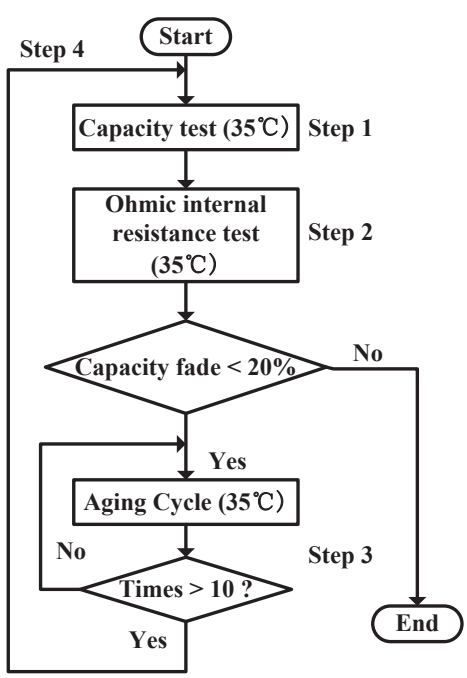

Fig. 2. The test procedure for the aging experiments.

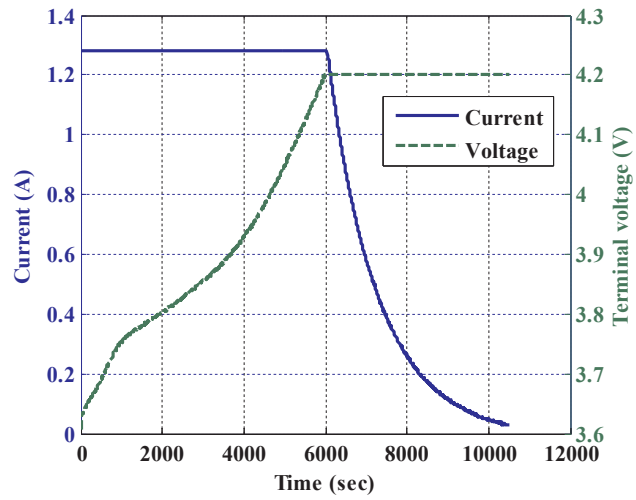

Fig. 3. Charge stages of lithium-ion battery.

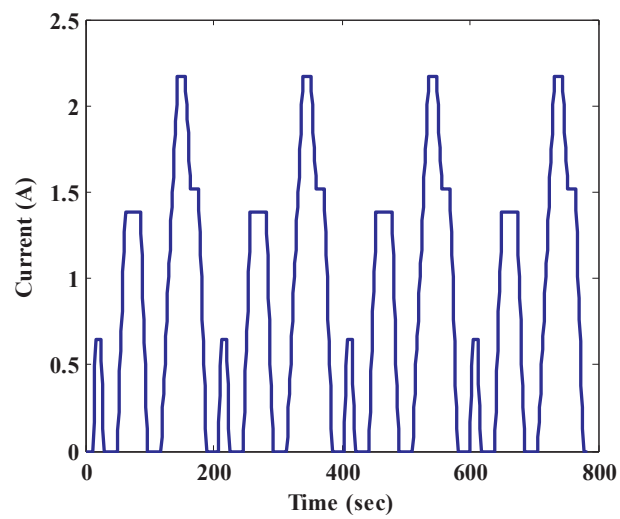

Fig. 4. Battery discharge current during one NEDC cycle.

$$C^i_{dec} = C^1_{cur} - C^i_{cur} \tag{1}$$

Ohmic internal resistance test: The battery is discharged with the NEDC (Fig. 4) until the discharge cutoff voltage reaches 2.75 V again after the standard charge. Different from current pulse profile, NEDC is one of the typical dynamic conditions to simulate the dynamic automobile travel process. And the load current of NEDC is slowly changing rather than mutating. The discharge capacity, voltage, and current are recorded simultaneously by IT8511+.

The real-time battery model parameters can be deduced based on the measurement data using RLS which will be explained in detail in Section 3.1. The mean value of the ohmic internal resistance during the SoC range from 80% to 30% is calculated as R_{cur} according to Eq. (2). One of the reasons is that the batteries used for EV applications are almost completely depleted (typically 95–25% SoC) and then charged using external power sources [26]. We used a more conservative SoC range (80–30%) for the ohmic internal resistance. And the initial average ohmic internal resistance is defined as R^1_{cur}, and the ith average ohmic internal resistance is R^i_{cur}. And the ith increase of ohmic internal resistance (R^i_{dec}) is defined as follows:

$$R_{cur} = mean\left(\sum_{SoC=30\%}^{SoC=80\%} \left(R_{0(SoC=30\%)} + \cdots + R_{0(SoC=80\%)}\right)\right)$$
$$R^i_{dec} = R^1_{cur} - R^i_{cur} \tag{2}$$

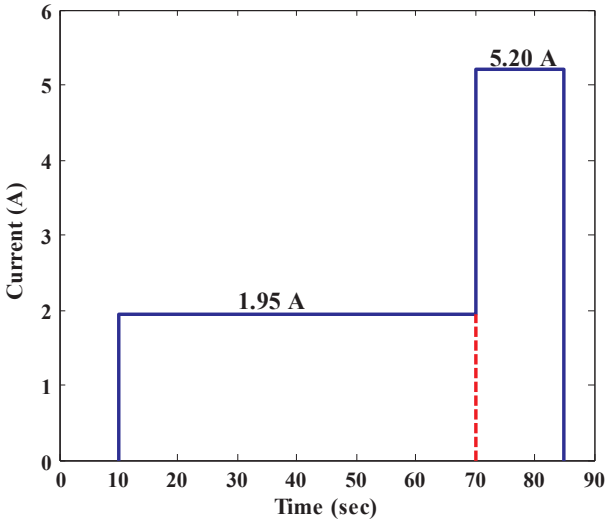

Fig. 5. Battery discharge current of sample No. 30.

Here, $R_{0(SoC)}$ represents the ohmic internal resistance for a certain SoC, and mean is a symbol for averaging. It's perhaps important to point out that the SoC is calculated by Coulomb-counting method based on its definition [1,27].

Aging cycle: Gu et al. [28] analyzed the effect factors of battery aging process, high temperature, high-rate discharge, depth of charge and pulse discharging are conducive to accelerate the aging process. Therefore, the temperature of the environment is set to 35 °C and the pulse discharge current for accelerating aging. We charge the battery fully, then discharge the battery with a pulse current (as shown in Fig. 5) until the discharge cutoff voltage reaches 2.75 V again. This step is repeated ten times as an integrated aging cycle.

The steps above are repeated continuously to investigate the characteristics of the battery until the battery capacity decreases by 20% of the nominal capacity.

Taking into account the impact of different temperatures on battery aging, the experiments of sample No. 17 are performed at different temperature with the same procedure of sample No. 30. However, it is necessary to point out that the capacity test (Step 1) and ohmic internal resistance test (Step 2) are at 25 °C, and the aging cycle (Step 3) is at 45 °C and the discharge current is composed of UDDS (Urban Dynamometer Driving Schedule) and constant current (as shown in Fig. 6).

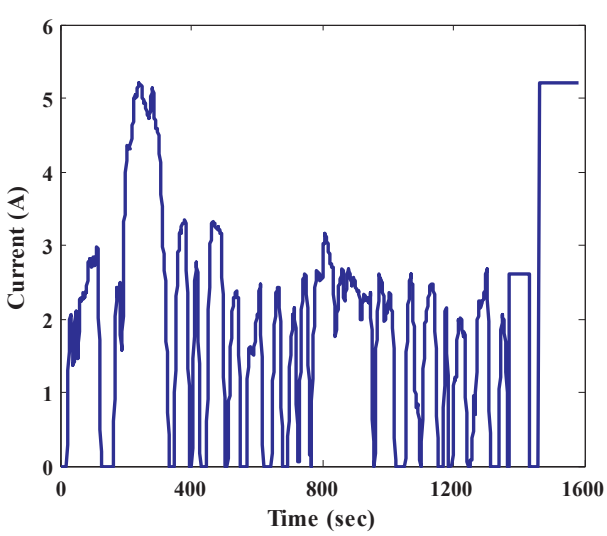

Fig. 6. Battery discharge current of sample No. 17.

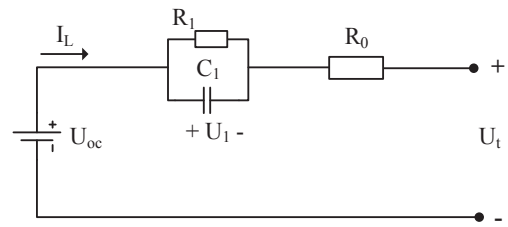

Fig. 7. Schematic diagram of Thevenin model.

3. Method

In this section, explanation of the battery model and parameters identification method are conducted firstly. Then, the correlation between the increase of ohmic internal resistance (R_{inc}) and the decrease of the capacity (C_{dec}) for each aging experiment is analyzed. Finally, a new SoH estimation is developed based on this correlation. The current ohmic internal resistance (R_{cur}) is an average value, which is the mean resistance based on RLS and the Thevenin model during the SoC range between 80% and 30%. The present available maximum capacity (C_{cur}) is measured for each aging experiment.

3.1. Battery model and parameters identification

Thevenin model is considered the best choice based on model complexity, accuracy, and robustness [29]. The schematic diagram of Thevenin model is shown in Fig. 7, and the electrical behavior can also be written as follows:

$$\begin{cases} U_{oc} = U_t + I_L R_0 + U_1 \\ I_L = \frac{U_1}{R_1} + C_1 \frac{dU_1}{dt} \end{cases} \tag{3}$$

where U_{oc} denotes the OCV and I_L represents the load current. R_1 and C_1 are, respectively, the polarization resistance and polarization capacitance of the RC network, R_0 is the ohmic internal resistance, U_t is the terminal voltage and U_1 is the voltage across the parallel RC network.

The accurate identification of the parameters for the battery models is very important as well [27]. With some modifications and improvements, the frequently-used recursive least squares (RLS) method can be used to optimize the system model. The problem of a standard RLS is that it usually results in the saturation phenomenon. This disadvantage is owing to the exponential growth of the covariance matrix. Therefore, based on the standard RLS method, an optimal forgetting factor is employed to give less weight to old data and more weight to recent data, the parameter estimates are updated at every sample interval on the basis of current and terminal voltage measurement at time k and previous estimates at time (k − 1) [30]. The system identification is realized as follows:

$$\begin{cases} e(k) = U_t(k) - \mathbf{H}(k)\theta(k-1) \\ \mathbf{K}(k) = \frac{\mathbf{P}(k-1) * \mathbf{H}^T(k)}{\mathbf{H}^T(k) * \mathbf{P}(k-1) * \mathbf{H}(k) + \lambda} \\ \mathbf{P}(k) = \frac{\mathbf{P}(k-1) - \mathbf{K}(k)\mathbf{H}^T(k)\mathbf{P}(k-1)}{\lambda} \\ \theta(k) = \theta(k-1) + \mathbf{K}(k)\mathbf{e}(k) \end{cases} \tag{4}$$

where $\mathbf{e}(k)$ is the prediction error of the battery terminal voltage. $\mathbf{H}(k)$ is the matrix of systematic data and $\theta(k)$ is the matrix of systematic parameters. $\mathbf{H}(k)$ and $\theta(k)$ are defined and preset [30]. $\mathbf{K}(k)$ is the algorithm gain and $\mathbf{P}(k)$ is the covariance matrix at the moment k, λ is the forgetting factor, typically $\lambda \in [0.95\ 1]$.

The schematic diagram for the online identification of the Thevenin model parameters is shown in Fig. 8. The initial values of the parameter vector $\theta(1)$ and its error covariance matrix $P(1)$ are preset. Based on load current $I_L(k)$, and terminal voltage $U_t(k)$ which are measured, the values of $\theta(k)$ at each moment can be adjusted adaptively.

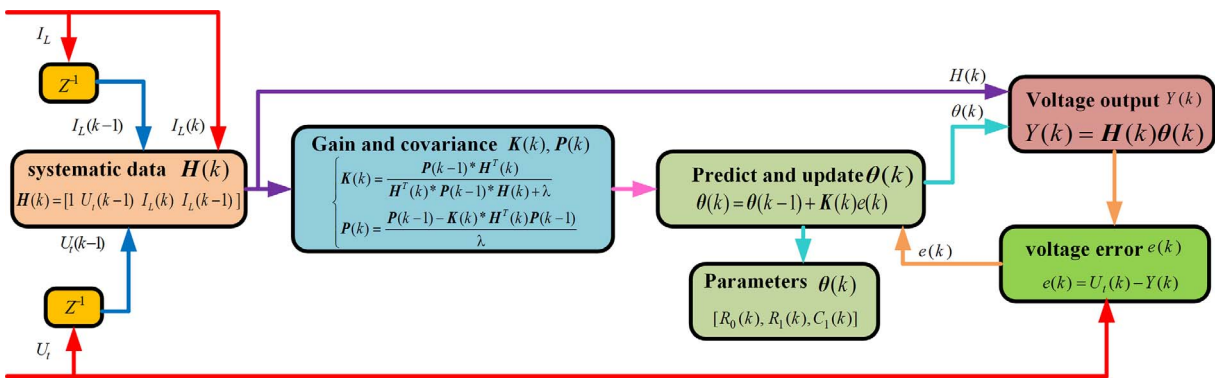

Fig. 8. Scheme of the identification of the Thevenin model parameters.

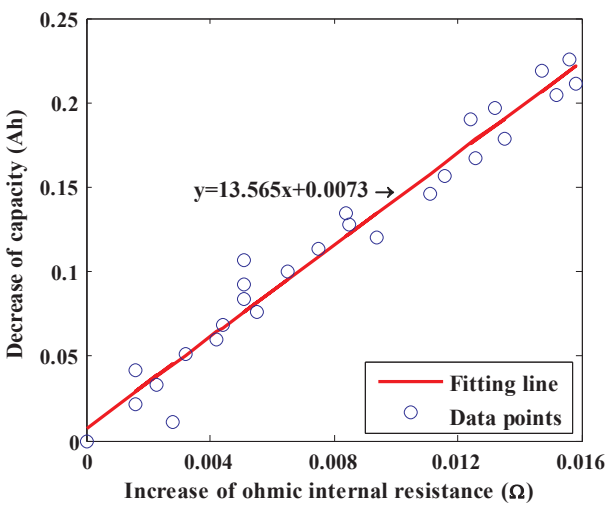

Fig. 9. Scatter diagram and fitting line.

3.2. Correlation analysis

To quantitatively evaluate the correlation between R_{inc} and C_{dec}, a series of aging experiments were implemented and we acquired a lot of experimental data, such as R_{cur} and C_{cur}. Both R_{inc} and C_{dec} are calculated using processing technology, and then a scatter diagram (Fig. 9) was created to describe the general relationship between them. The data for correlation analysis are from sample No. 30, and they are handled with a low-pass filter in advance. This scatter diagram can help us to decide whether the relationship between them is a linear correlation or not.

In Fig. 9, with the increase of R_{inc}, C_{dec} increased gradually, which indicates that there is a positive linear relationship between them. Therefore, the linear least squares fit method was implemented to reflect their dependency - see the solid line in Fig. 9, which can also be expressed as an equation. However, this is only information which is intuitively obtained from Fig. 9, more relevant parameters are required to elaborate their relationship.

In order to quantitatively evaluate the degree of correlation, three common correlation coefficients, Pearson's correlation r_P, Kendall tau rank correlation r_K, and Spearman's rank correlation r_S, were calculated and used – see Table 2. Clearly, there is a linear relationship between them.

The Pearson correlation analysis is used to quantitatively validate the linear relationship between variables R_{inc} and C_{dec}. It is between -1 and $+1$, the greater absolute value of r_P represents more perfect linear relationship exists between the variables R_{inc} and C_{dec}. If it equals 0, no correlation exists between them. Kendall tau rank correlation coefficient is a non-parametric statistic used to measure the degree of correspondence between two rankings and assessing the significance of this correspondence. For all other arrangements the value lies between -1 and 1, and increasing values imply an increasing agreement between the rankings. If the rankings are completely independent, the coefficient has the value 0 on average. In addition, Spearman rank correlation analysis is a nonparametric way to assess the strict monotonic relationship between two series. It is also between -1 and $+1$, If $r_S > 0$, there is a strong positive correlation between R_{inc} and C_{dec}, otherwise there is a negative correlation. From the Table 2, the conclusion can be drawn that there is a strong linear correlation between R_{inc} and C_{dec} in both qualitative and quantitative ways.

Table 2
The statistic list of correlation coefficients.

Correlation coefficient	Value
Pearson's correlation coefficient	0.9818
Kendall tau rank correlation	0.8969
Spearman's rank correlation coefficient	0.9806

3.3. SoH estimation

Based on the correlation analysis above, the SoH estimation is proposed, which is suitable to estimate SoH online.

3.3.1. General definition of SoH

Usually, SoH is 100% when the battery is new and 0% when the maximum capacity of the battery decreases to a certain minimum value. A commonly used definition is that when the C_{dec} is up to 20% of the nominal capacity (C_{new}) the battery is at the end of its lifetime. The definition of SoH_c is shown in Eq. (5) [24].

$$SoH_c = \left(1-\frac{C_{new}-C_{cur}}{0.2\times C_{new}}\right)\times 100\% \tag{5}$$

The method to calculate the SoH above is based on capacity. In addition, the ohmic internal resistance can be seen as another definition of SoH. Eq. (6) is the definition of SoH_R [1]:

$$SoH_R = \frac{R_{eoL}-R_{cur}}{R_{eoL}-R_{new}}\times 100\% \tag{6}$$

where the R_{cur} is the current average ohmic internal resistance mentioned in Eq. (2), R_{eoL} is the ohmic internal resistance at the end of the lifetime, R_{new} is the ohmic internal resistance at the start of the lifetime. As a result, if R_{new}, R_{eoL} and R_{cur} can be estimated accurately, the SoH_R can be determined with enough accuracy using the definition.

3.3.2. The proposed SoH estimation

Based on the relationship between the ohmic internal resistance and capacity, as mentioned above, the linking equations Eq. (7) can be

formulated:

$$\begin{cases} SoH_c^m = SoH_R^m = \frac{\hat{R}_{eol} - R_{cur}^m}{\hat{R}_{eol} - \hat{R}_{new}} \times 100\% \\ SoH_c^n = SoH_R^n = \frac{\hat{R}_{eol} - R_{cur}^n}{\hat{R}_{eol} - \hat{R}_{new}} \times 100\% \end{cases} \tag{7}$$

where SoH_c is calculated by the available maximum capacity which is measured offline, SoH_R is calculated using the ohmic internal resistance. R_{cur}^m and R_{cur}^n are the average ohmic internal resistances (in the SoC range between 80% and 30%) of two random different aging stages, respectively.

At the same aging stages, the SoH estimation results are equal, whichever definition we use. Therefore, in this binary linear equation group, SoH_c^m, SoH_c^n, R_{cur}^m and R_{cur}^n are known variables, while $\hat{R}_{new}$ and $\hat{R}_{eol}$ are unknown variables. Then, we can solve the equations, and the estimated values ($\hat{R}_{new}$ and $\hat{R}_{eol}$) can be obtained. Usually, it takes a very long time to age the battery to obtain R_{eol}. Moreover, if the battery has already been used for some time, it is impossible to acquire R_{new} again. The proposed method in Eq. (4) does not require any particular ohmic internal resistance tests to acquire R_{new} and R_{eol} directly, which is ideal for online SoH estimation.

To better describe the proposed method, a schematic diagram is drawn – see Fig. 10.

SoH_R^i can be estimated using $\hat{R}_{new}$ and $\hat{R}_{eol}$ in Eq. (4), and the SoH is then determined using Eq. (8):

$$SoH_R^i = \frac{\hat{R}_{eol} - R_{cur}^i}{\hat{R}_{eol} - \hat{R}_{new}} \times 100\% \tag{8}$$

In addition, the statistical analysis in Section 3.1 demonstrates the feasibility of SoH estimation based on the relationship between R_{inc} and C_{dec} .

Furthermore, according to the datasheet and definition, C_{new}, R_{new} and R_{eoL} are acquired in tests at 35 °C. Hence, C_{cur} and R_{cur} should be taken at the same temperature too. However, in the actual application, the temperature could have been any value. Therefore, the values for C_{cur} and R_{cur} should be converted to the reference temperature from the actual test temperature. Follow-up studies of this part are being planned.

In summary, an online general framework for the new estimation SoH is presented in Fig. 11.

Thus, the battery SoH can be estimated in accordance with the operating process as follows:

Step 1: At first, the Thevenin model of battery is constructed, and the RLS is applied to identify the model parameters. The identified ohmic internal resistance is considered to be the true value of ohmic internal resistance, then the average ohmic internal resistances during the SoC range from 80% to 30% are calculated;
Step 2: The average ohmic internal resistances (R_{cur}^m and R_{cur}^n) and capacities (C_{cur}^m and C_{cur}^n) at two different aging states are chosen to solve the $\hat{R}_{new}$ and $\hat{R}_{eol}$ in Eq. (7); Thus, the SoH estimation method based on ohmic internal resistance is established – see Eq. (8).
Step 3: The current average ohmic internal resistance R_{cur}^i at the other aging state can also be acquired based on ***Step 1***; based on the proposed estimation method in ***Step 2***, and the current SoH can be estimated online.

In the following, the proposed method is verified using the mass aging experimental data, which are obtained through a series of aging experiments at different temperatures in Section 2.2.

4. Results and discussions

In this section, we first verify the accuracy and robustness of the battery model and the identification algorithm and then discuss the effectiveness and accuracy of the proposed method.

4.1. Evaluation of the Thevenin model and RLS

In order to guarantee the estimation accuracy for SoH, using the ohmic internal resistance, the Thevenin model and RLS should reflect the dynamic characteristics of the battery accurately. A comparison of the terminal voltage via an NEDC test between the model-based simulation and the measured terminal voltage is implemented. The comparison profiles are shown in Fig. 12. The estimated terminal voltage tracks the references accurately and the identification method performs well.

The terminal-voltage estimation error is shown in Fig. 13. The results suggest that the error at the initial stage and the ending stage increase sharply because RLS requires some time to converge at the initial stage. However, the error between the battery SoC values between 80% and 30% is no larger than 5 mV (Fig. 13). This indicates that

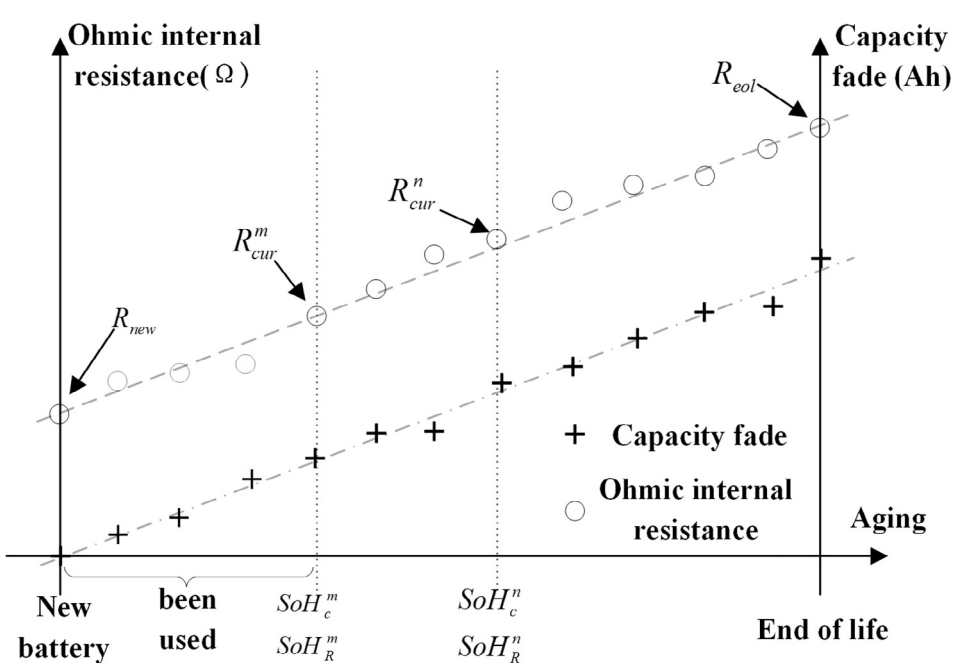

Fig. 10. Schematic diagram of the proposed method.

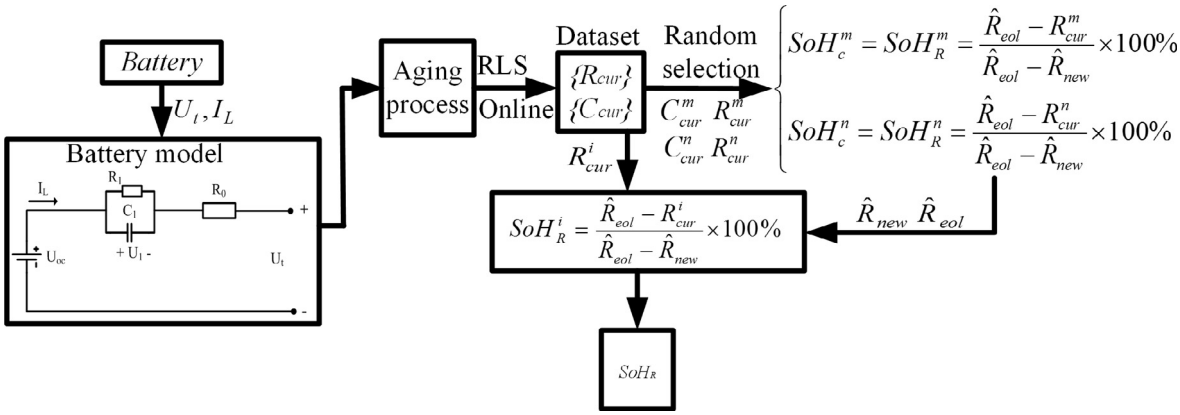

Fig. 11. The implementation flowchart of the new SoH estimation.

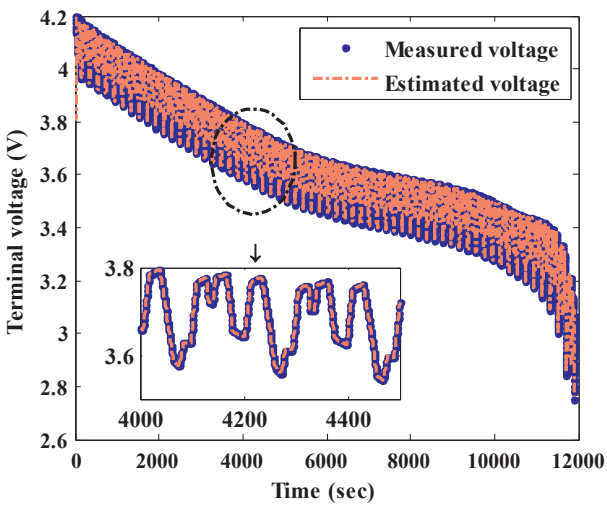

Fig. 12. The terminal voltage profiles of the experiment and the estimates.

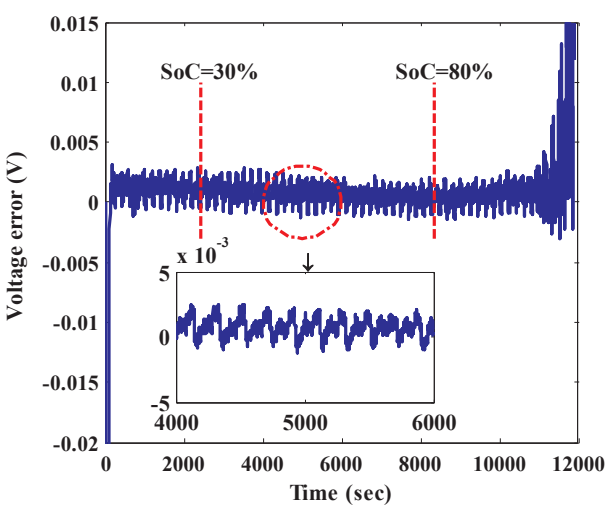

Fig. 13. The terminal voltage error between experiments and estimates.

the Thevenin model and RLS are sufficiently reliable and accurate to track the dynamics of the battery. Not all data are required during one cycle to average the ohmic internal resistance. In practice, it is better to utilize the data for which SoC is between 80% and 30%.

Through the comparison of measured and estimated voltage, the accuracy of the Thevenin model and RLS has been verified. Furthermore, the battery ohmic internal resistance values can be

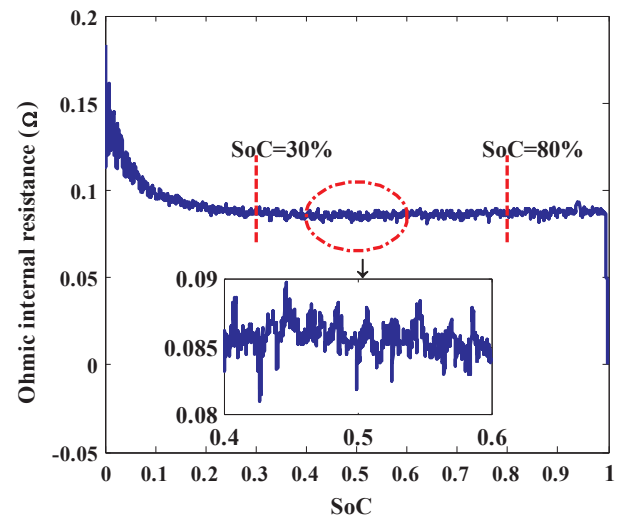

Fig. 14. The ohmic internal resistance curve.

obtained as shown in Fig. 14, which show little fluctuation in the SoC range between 80% and 30%. Then, the parameters of the model will be used to estimate SoH.

Throughout the whole discharging process, the ohmic internal resistance changes significantly, and the ohmic resistance increases rapidly at the end of the discharge process. However, when the source is in the range between 80% and 30%, the battery ohms internal resistance value is almost smooth, with small local variations. As shown in Fig. 14, the ohmic internal resistances are obtained in the SoC range between 80% and 30%, and the mean value of the resistance is calculated using Eq. (2) as $R_{cur} = 0.086$. This steady value provides a base for the SoH estimate using the Thevenin model and RLS.

When the EV battery experiences partial discharge/charge cycles, the range of SoC may not be between 80% and 30% exactly. However, the ohmic internal resistances remained essentially unchanged when SoC is greater than 30% according to Fig. 14. In order to elaborate that different SoC ranges have little effect on average internal resistance, several dynamic condition cycles of sample No. 30 are selected randomly, and the average resistances in some elective subintervals are calculated as shown in Table 3.

Table 3 shows that, there is little variation for the average resistances when the SoC is greater than 30%. On the whole, the differences of average resistance are no more than 1 mΩ in different ranges of SoC. In other words, the average resistances in other ranges of SoC can be approximatively equivalent to the average resistance in the SoC range between 80% and 30%. The invariance of average internal resistance also contributes to the online implementation of the estimation

Table 3
The values of average ohmic internal resistances under different ranges of SoC.

Ranges of SoC	3rd[a] average ohmic internal resistances	6th average ohmic internal resistances	17th average ohmic internal resistances
[30–80%]	0.0812	0.0824	0.0895
[40–80%]	0.0812	0.0824	0.0894
[50–80%]	0.0813	0.0826	0.0895
[60–80%]	0.0815	0.0828	0.0899
[70–80%]	0.0815	0.0829	0.0903
[80–90%]	0.0816	0.0827	0.0910

[a] 3rd is the result based on the third ohmic internal resistance test.

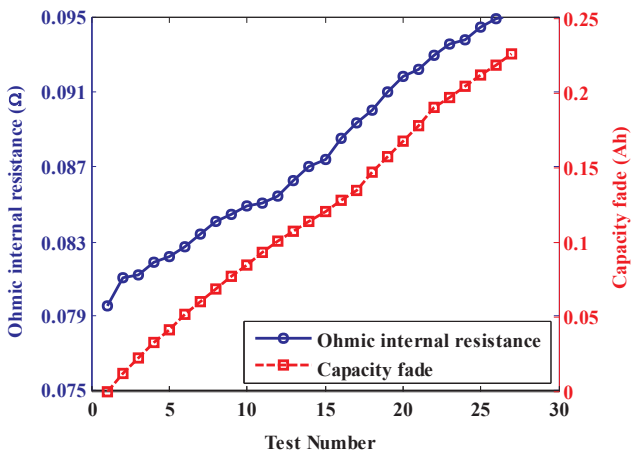

Fig. 15. The results of the proposed method.

method.

The method was also verified by a series of aging experiments. The relationship between capacity fade and ohmic internal resistances turned out to be suitable to estimate SoH. The results of battery No. 30 are shown in Fig. 15 and are similar to Fig. 10. There is a linear relationship between ohmic internal resistance and capacity fade, and this relationship can be used to estimate SoH according to this study.

4.2. Experiment results of SoH estimation

In the proposed method, the values for capacity and ohmic internal resistance at two different aging states are required to estimate R_{new} and R_{eol}. Hence, in a random aging cycle (not the first) experiment is chosen, and then the ohmic internal resistance R_{cur}^{m} and capacity C_{cur}^{m} are chosen and recorded. After several aging cycles, the ohmic resistance R_{cur}^{n} and capacity C_{cur}^{n} for the other aging states were chosen and recorded again. To avoid local errors producing global effects, the number of aging cycles needs to be as high as possible. Therefore, on the one hand, SoH_c^m and SoH_c^n can be calculated using the definition as the accurate values of battery SoH. On the other hand, SoH_R^m and SoH_R^n equal to SoH_c^m and SoH_c^n, respectively. Thus, $\widehat{R}_{new}$ and $\widehat{R}_{eol}$ can be obtained by solving the binary linear equation group – see Eq. (7).

For the battery No. 30, we can calculate $\widehat{R}_{new} = 0.0793\ \Omega$ and $\widehat{R}_{eol} = 0.1170\ \Omega$ using Eq. (7). The ohmic internal resistance of the actual measurement (R_{new}) is 0.0796 Ω. For the battery No. 17, we can also calculate $\widehat{R}_{new} = 0.0866\ \Omega$ and $\widehat{R}_{eol} = 0.1260\ \Omega$ respectively, and the ohmic internal resistance of the actual measurement (R_{new}) is 0.0864 Ω. The estimate errors for R_{new} are lower than 1 mΩ, and the high-precision estimates is very favorable to establish a highly sensitive method to estimate SoH.

Following the test schedule in Section 2.2, more tests have been done to verify the estimation accuracy of SoH. Based on the new proposed method, the SoH_R of sample No. 30 and sample No. 17 can be estimated using Eq. (8), while the actual SoH_c values were also calculated using Eq. (5). The estimated results are plotted in Fig. 16.

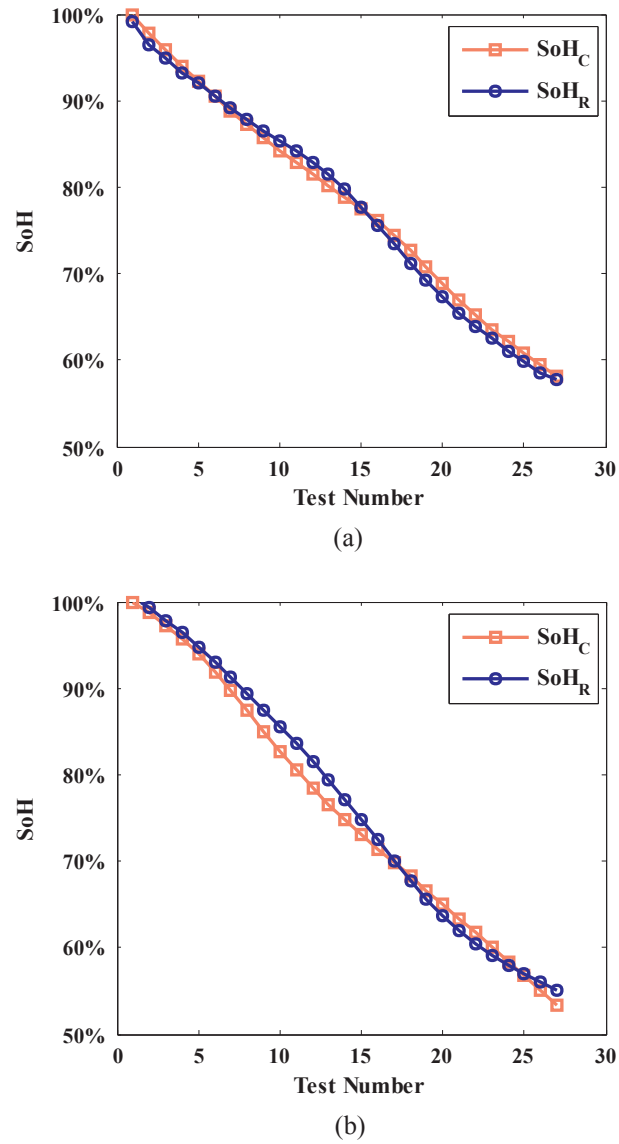

Fig. 16. The SoH estimation results (a) No. 30 battery (b) No. 17 battery.

In order to discuss both the accuracy and reliability of the new method, the estimation error was calculated according to Eq. (9) with the final calculated values of SoH_R and SoH_c. In addition, the error profiles for the SoH estimation are shown in Fig. 17. To evaluate the estimation accuracy in detail, we use the mean absolute error (MAE), root-mean-square error (RMSE), and variance to evaluate the error performance. A detailed statistical analysis is shown in Table 4.

$$Error = SoH_R - SoH_c \tag{9}$$

The application of the new method on the testing data from the laboratory indicates that the estimated SoH tracks the references accurately (Fig. 16), and the maximum error of the estimation is less than 4.0% (Fig. 17 and Table 4) for each battery. The results of this work also suggest that the proposed method can deliver an acceptable accuracy for SoH estimation and that it is effective and suitable to implement online.

The experiment results from this work suggest that the capacity fade are related to the increase of the ohmic internal resistance over aging, since they follow the same growth trend. Such a linear correlation also implies that they are most likely attributed to the same aging mechanism. On the one hand, Solid Electrolyte Interface (SEI) formation

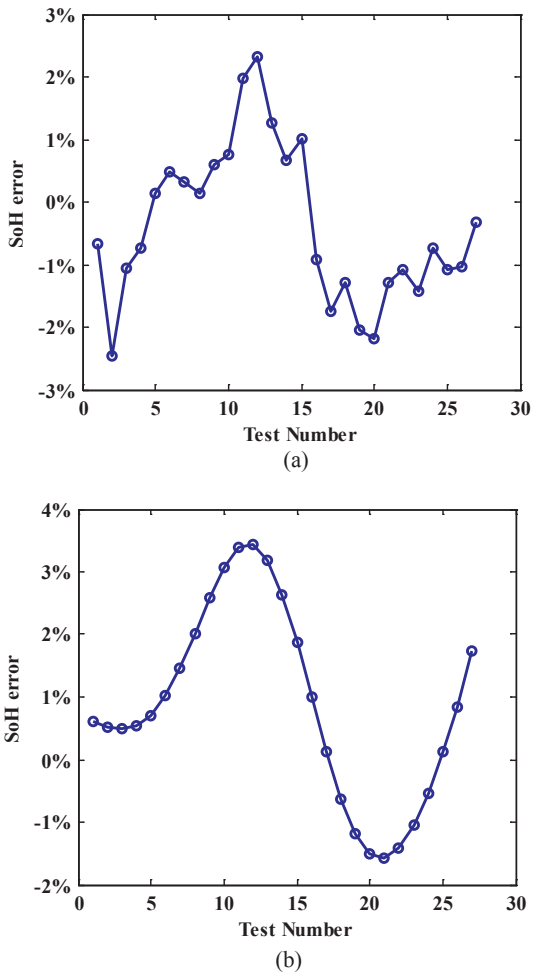

Fig. 17. Estimation errors of SoH (a) No. 30 battery (b) No. 17 battery.

Table 4
The statistic list of SoH errors.

Number	Maximum	MAE	RMSE	Variance
No. 30 battery	2.45%	0.0038	0.0128	1.55×10^{-4}
No. 17 battery	3.76%	0.0145	0.0176	2.43×10^{-4}

can cause a significant increase of the impedance. At the same time, the number of lithium-ion which are one of the active materials would reduce. Thus, the battery capacity decreases and the battery impedance increases as a consequence. Therefore, a new method to determine SoH that uses the correlation between the decrease of capacity and the increase of the impedance is proposed. This method can be considered as an attempt to define SoH such that it could reflect both the change in battery impedance and capacity. In all comparisons it can be observed that the SoH estimates indicate that there is some correlations between the increment of ohmic internal resistance and capacity fade. The capacity fade can be determined to some extent by considering the increment of ohmic internal resistance. This proposed method is an example to define the SoH, which could represent both the degeneration caused by ohmic internal resistance increase and capacity fading. The application of these adaptive and robust algorithms will further improve the accuracy of the SoH estimation. This will be our future research goal.

5. Conclusions and future work

Due to the irreversible aging process in lithium-ion batteries, it is difficult to obtain the ohmic internal resistance at both the start and the end of its lifetime. This difficulty limits the usefulness of ohmic internal resistance to estimate SoH. For this reason, a new SoH estimation method is proposed and verified experimentally. Our main results are as follows:

(1) Based on RLS and the Thevenin model, the ohmic internal resistance is identified for dynamic loading profiles. Then, the mean of the ohmic internal resistance during the SoC range between 80% and 30% is calculated. This value is a good basis for an accurate SoH estimation.
(2) The ohmic internal resistances at the start and end of the battery lifetime are estimated via two discharging tests that use the correlation between ohmic internal resistance and capacity. Then, the linking equation for the SoH estimation is built. The SoH estimation does not require prior knowledge of the ohmic internal resistance. Yet, it is of acceptable accuracy with maximum estimation errors of less than 4.0% during the aging experiments of the lithium-ion battery.
(3) In addition, the new method was also verified with measurement data from the aging experiments. The method requires very little computation power. Hence, the proposed method is suitable to estimate SoH online in EVs.

Future work could focus on building a better model for the SoH estimation, which considers the SoC and state-of-power (SoP) combined with other identification algorithms. This would permit the implementation of accurate and efficient online identification of the model parameters. Furthermore, one would obtain perfect ohmic internal resistance characteristics using experimental data (at different temperatures) for use in EVs.

Acknowledgements

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